

Report 01

Field Research & Stakeholder Survey

Project:	JalJeev (formerly ORCA) — Agentic Marine Intelligence System
Phase:	Pre-Development / Discovery
Period:	July–August 2026
Authors:	Team Felix, KIIT University, Bhubaneswar
Problem Statement:	SIH26176 (ISRO)

1. Objective

Before writing a single line of code, Team Felix set out to understand the real decision-making environment of coastal fishing communities. The ISRO problem statement (SIH26176) called for an *"agentic marine intelligence system"* — but agentic for whom? And solving what, exactly?

We needed to answer that question not from a conference room in Bhubaneswar, but from the harbours, beaches, and landing centres where fishermen actually make their daily go/no-go decisions. This report documents the field research that shaped every subsequent product decision — beginning with an unexpected opportunity in Sri Lanka, and continuing with structured interviews along the Odisha coast at Puri.

2. Research Methodology

2.1 Phase A — Opportunistic Field Interview: Sri Lanka

One of our team members travelled to Sri Lanka for personal reasons and seized the opportunity to interview local fishermen in the Arugam Bay area — a well-known coastal region on Sri Lanka's southeastern coast. This was not a planned research trip; it was an opportunistic but rigorous field interview that became the single most important input to our product design.

Why this mattered: Arugam Bay is a relatively developed, internationally known beach destination. If fishermen here — in a tourism-adjacent, connected area — were still operating without digital tools, then the problem we were trying to solve was not hypothetical. It was real and urgent.

2.2 Phase B — Structured Surveys: Puri Coastal Region, Odisha

Following the Sri Lanka insights, we conducted a series of structured interviews with artisanal fishermen at the **Puri coastal region in Odisha** — the primary deployment target for JalJeev. These interviews validated and extended the Sri Lanka findings within an Indian context, and directly informed the product's vessel profiles, risk thresholds, language requirements, and offline architecture.

2.3 Additional Stakeholder Outreach

Stakeholder	Method	Locations
Mechanized Trawler Operators (Seizure)	Semi-structured interviews	Visakhapatnam (AP); Kochi (Kerala)
Coastal Sailors & Dhow Operators	Phone interviews	Gujarat coast; Konkan coast
Disaster Management Officers	Structured questionnaire, video call	CADMA (Odisha); Andhra Pradesh DM authority
Marine Researchers	Academic interviews	CMFRI Kochi; INCOIS Hyderabad (secondary)

3. The Sri Lanka Interview — Arugam Bay

3.1 Interview Details

Location	Arugam Bay, including Pottuvil Point, Baby Point and Whiskey Point
Interviewee	Local fisherman
Fishing experience	Approximately 20 years
Approximate boats operating	Approximately 100 boats

3.2 Interview Findings

Q1: Trip Process

The fishermen typically leave around 3:00 AM and return between 7:30–8:00 AM after fishing for 4–5 hours. Their process is based on experience and direct observation of the sea. Once at the fishing area, they look for signs of fish and move to another location if fish aren't found.

Q2: Location Decision

Fishermen rely on experience and observation of the water. They don't stay at one fixed location — they move between multiple fishing areas based on where they believe fish are available. Traditional signs and communication between experienced fishermen also inform their decisions.

Q3: Problems & Dangers

Major difficulties: high tides, large waves, and rain. Rough conditions make it difficult for the boat to move safely. Fishing is season-dependent, with June–September being the most important period. Outside the main season, sea conditions make fishing more difficult.

Q4: Navigation & Failures

Fishermen do not rely on GPS or mobile navigation as part of their normal process. Instead, they depend on experience and familiar landmarks. The light tower/light from the coast is an important navigation method for returning or identifying shore direction at night. Flashlights are used to observe the water and look for fish.

Q5: Decision Triggers

If they don't find fish, they move to another location. They stop or return when sea conditions become unsafe, particularly with strong waves or high tides. These decisions are made dynamically based on what they observe while at sea.

Q6: Digital Tools

The fisherman does not use weather apps, mobile alerts, fishing applications, PFZ information, or similar digital tools. No weather or fishing-condition alerts are received before or during the trip. **This is particularly notable because Arugam Bay is a relatively developed, well-known beach/tourist area.**

3.3 Key Findings Summary

Category	Finding
Departure & Return	3:00 AM departure, 7:30–8:00 AM return
Trip duration	~4–5 hours
Location selection	Experience + observation of water
Relocation trigger	Not finding fish
Stop/return trigger	High waves, high tide, rain, unsafe conditions
Navigation	Experience + visual landmarks + coastal light tower
Boats in area	~100 boats
Weather apps	Not used
Mobile alerts	Not used
Fishing apps/PFZ	Not used
Fishing experience	~20 years

4. The Design Insight

The Sri Lanka interview revealed something critical: the decision-making framework is not:

Experience → Weather app → Real-time decision

but rather:

20 years of experience → direct observation → immediate action

The fisherman is operating with ~100 boats in the same area. So the important question for the project is **not** "Can we make fishermen use an app?" The more meaningful problem is:

"How can existing, experience-based fishing decisions be supported with reliable real-time information — without replacing the fisherman's local knowledge?"

That distinction became central to the SIH presentation. We were not claiming that fishermen have no knowledge. Quite the opposite: **they have enormous practical knowledge**, but much of it is informal, local, and inaccessible to others in real time.

JalJeev's job is not to replace that knowledge. It is to **augment it** — to give the fisherman who already knows his waters access to the satellite, ocean model, and weather forecast data that currently sits on government portals he will never visit, in a language (English) he does not read, in a format (NetCDF) he cannot open.

5. Puri Coastal Region Surveys — Indian Context Validation

5.1 Why Puri

Following the Sri Lanka interview, we needed to validate the findings in our primary deployment context: the Indian coast. Puri, Odisha was chosen because:

- It is a major artisanal fishing hub on India's eastern seaboard
- The Odisha coast faces significant seasonal cyclone risk (Bay of Bengal)
- Odia-speaking fishermen represent a core target user group for JalJeev
- KIIT University's proximity enabled direct field access

5.2 Interview Programme

We conducted semi-structured interviews with fishermen at the Puri landing centres and harbour areas. The interviews followed the same 7-question protocol used in Sri Lanka, adapted for the Indian context with additional questions about INCOIS awareness, cyclone experience, language preference, phone usage, and vessel specifics.

5.3 Puri Findings — Consistent with Sri Lanka

Finding	Sri Lanka	Puri, Odisha
Primary decision method	Experience + visual observation	Experience + visual observation
Use of weather apps	Not used	Rarely — generic weather only
Use of PFZ/INCOIS data	Not used	Never heard of it
Navigation at sea	Landmarks + light tower	Landmarks + experience
Cyclone awareness source	Seasonal knowledge	TV news, village loudspeaker
Language preference	Sinhala/Tamil	Odia — strongly preferred
Smartphone ownership	Mixed	~80% had basic Android phones
Data connectivity at sea	None mentioned	Intermittent, lost beyond ~15 km
Voice preference	Not tested	~70% preferred voice over typing

5.4 Additional Puri-Specific Findings

- No fisherman we interviewed had ever visited the INCOIS website. Not one.
- Cyclone warnings reached fishermen via village panchayat loudspeakers and TV news — not via any digital channel. Sometimes too late.
- Several fishermen mentioned losing colleagues to capsizing in sudden squalls and unexpected wave conditions — the exact scenario JalJeev's hard-constraint vetoes are designed to prevent.
- Vessel profile data collected: Typical Puri artisanal boat is 6–8 metres, 15–40 HP outboard, crew of 2–4, max safe wave ~1.5–2.0 m, operational range 40–60 km — these numbers directly informed the vessel class architecture.

6. Gap Analysis — Survey-Validated

6.1 India's marine data infrastructure is world-class — but completely inaccessible

System	Operator	Output	Accessibility to Fishermen
Ocean State Forecast (OSF)	INCOIS	5–7 day wave/current/SST forecast	English only, scientific format, never seen by any interviewee
PFZ Advisory	ISRO/INCOIS	Fishing zone maps	Published as maps on portal — no fisherman had heard of it
Cyclone Track + Warnings	IMD	Track, windfield, coastal flooding	Reaches fishermen via TV/loudspeaker, not personalised
EOS-06/OCM-3 Chlorophyll	ISRO MOSDAC	Satellite ocean colour	Requires account, NetCDF literacy — completely alien
EEZ/MPA Boundaries	Marine Regions / WPA	Geofencing	Not available to fishermen in any form

6.2 The Technology Gap

Existing Approach	Limitation Confirmed by Field Research
Single weather app	No ocean data, no PFZ, no geofencing, no vessel context
INCOIS OSF portal	No fisherman interviewed had ever seen it
Generic LLM chatbot	Would hallucinate marine data; no real data, English only
Simple chatbot + weather API	No geofencing, no risk engine, no offline mode

7. Design Implications — Survey-to-Requirement Mapping

Survey Finding	Product Requirement
Fishermen rely on experience, not technology	Augment, don't replace — support existing knowledge with real-time data
Zero use of PFZ, INCOIS, weather apps	System must deliver data in a form fishermen will actually use
Voice preferred over typing (~70% in Puri)	Voice input/output mandatory
Regional languages only	10+ Indian language support required
Night navigation uses light towers	System must work in low-light, minimal-interaction mode
Decision to move/return is at-sea observation	System must provide proactive alerts before conditions deteriorate
Cyclone warnings arrive too late	Push alerts directly to fisherman's phone
Colleagues lost to capsizing	Hard safety vetoes that cannot be overridden
Connectivity lost beyond ~15 km	Offline mode with cached forecasts and local risk rules
~100 boats in same area	Fleet-level intelligence is a future opportunity

8. Competitive Landscape Assessment

Solution	Weakness (Validated by Survey)
Windy.app	No PFZ, no Indian language, no vessel-aware risk
INCOIS mobile app	Data display only, no decision engine, English only
Fisher Friend (Tamil Nadu)	Limited to TN, SMS-only, no real-time risk scoring
Generic weather apps	No ocean state, no geofencing, no safety scoring

9. The Central Design Principle

The Sri Lanka and Puri interviews produced a single guiding principle that drove every subsequent architectural decision:

JalJeev does not replace the fisherman's 20 years of experience. It gives him access to the INCOIS forecast, the IMD cyclone track, the MOSDAC chlorophyll map, and the GEBCO bathymetry chart — translated into his language, spoken aloud, and delivered to the phone that is already in his pocket.

The architecture's innovation is not in replacing INCOIS or IMD. It is in building the **intelligent orchestration and decision layer** that can consume those authoritative models, compare them, quantify their uncertainty, and turn the resulting ensemble into explainable decisions for fishermen and maritime stakeholders.

10. Next Steps

- Translate field findings into formal Problem Statement Analysis and PRD
- Register for ISRO SIH26176 and form Team Felix
- Register for all data source APIs (MOSDAC, IMD, Copernicus, NASA Earthdata — several take 2–5 days)
- Begin architecture design with the persona-driven, survey-validated requirements as the specification

*Report prepared by Team Felix, KIIT University, Bhubaneswar
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